Name: YOUSSEF

Surname: SAMOUNEH

ID No: 19049929

Group: 5




$a=9 \quad b=2 \quad c=9 \quad Q=20\,N \quad L=192\,cm$

$\hat{\sigma}_{min} = -20\ N/mm^2 \qquad \sigma_{max} = 0$

$$M_x = -QL + P \cdot e_y = -20 \cdot 1920 + P \cdot e_y = -38400 + P e_y$$

$$I_x = \frac{150 \cdot 300^3}{12} = 3375 \times 10^5\ mm^4 \qquad N = -P$$

$$\sigma_z = \frac{-P}{150 \times 300} + \frac{-P \cdot e_y + 38400}{3375 \times 10^5}\, y \qquad \text{at } y = -150\,mm$$

$\sigma_{max} = 0$

$$-2{,}22 \times 10^{-5} P - 4{,}44 \times 10^{-7} P \cdot e_y - 0{,}017 = +20$$

at $y = 150$

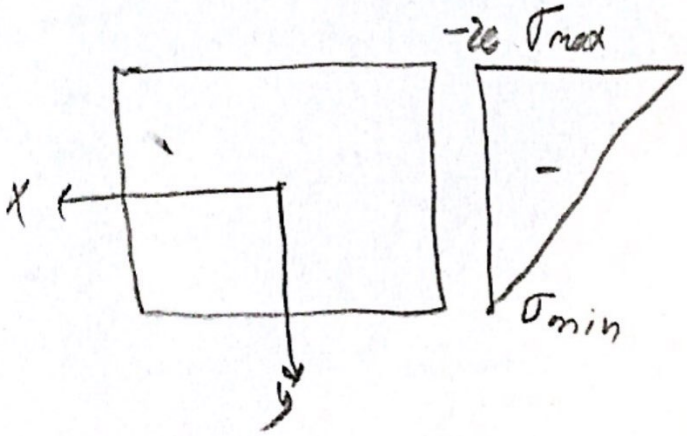
$$-2{,}22 \times 10^{-5} P + 4{,}44 \times 10^{-7} P e_y + 0{,}017 = 0 \qquad \sigma = -20$$

$$-4{,}44 \times 10^{-5} P = -20$$

$$\boxed{P = 4500\ N}$$

$$\boxed{e_y = 5\,cm}$$

b)




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